AI Training Material Database

Background & Description

The purpose of this project is to organize publicly available information on the internet to effectively train large language models (LLMs) in specific fields. Modern LLMs display a great breadth of knowledge across many subjects, allowing users to retrieve information without navigating large amounts of intermediary data. However, LLMs often lack sufficient knowledge in more niche areas of interest. This can result from factors such as data sparsity, level of embeddedness, and overall user demand. LLMs may also rely on outdated information, and this project provides a design that users can apply to update their own databases.

Fine-tuning an AI directly is expensive, slow, and difficult to maintain. For that reason, creating a database containing the desired information is a more efficient and reliable method for supporting niche knowledge. While modern LLMs rely on large datasets, they are heavily focused on general knowledge and become more error-prone as topics become specialized. Additionally, their data structures are often hidden due to security and proprietary concerns. This project proposes a standard way to organize LLM training material in a database where users can add knowledge from niche areas of academia and pop culture.

The database has seven entity tables from which relationships are formed, each contributing to the organization and structure of the system:

- The Users table contains information about users created through the sign-up page, allowing for user identification and management.
- The PDF_Resources table stores metadata for all uploaded PDF files, which may be public or private, and acts as the main source of training data.
- The AI_LLMs table contains information about chatbots created by users. These bots use uploaded resources to learn and fine-tune themselves according to their settings, and may also be public or private.
- There are four attribute tables: Authors, Subjects, Sources, and Status. These tables act as classifiers for the resources table, improving organization and searchability.

Furthermore, relationships define how data is connected within the database:

- The uploads relationship links the Users and PDF_Resources tables, identifying which user uploaded each resource.
- The creates relationship links the Users and AI_LLMs tables, identifying which user created each chatbot.
- The training relationship is a junction table between AI_LLMs and PDF_Resources that stores information about chatbot training.
- WrittenBy, Categorizes, Originates, and MarkedAs are relationships between the attribute tables and PDF_Resources where the first two form the corresponding junction tables and the last two form the attribute ID columns.

ER Diagram

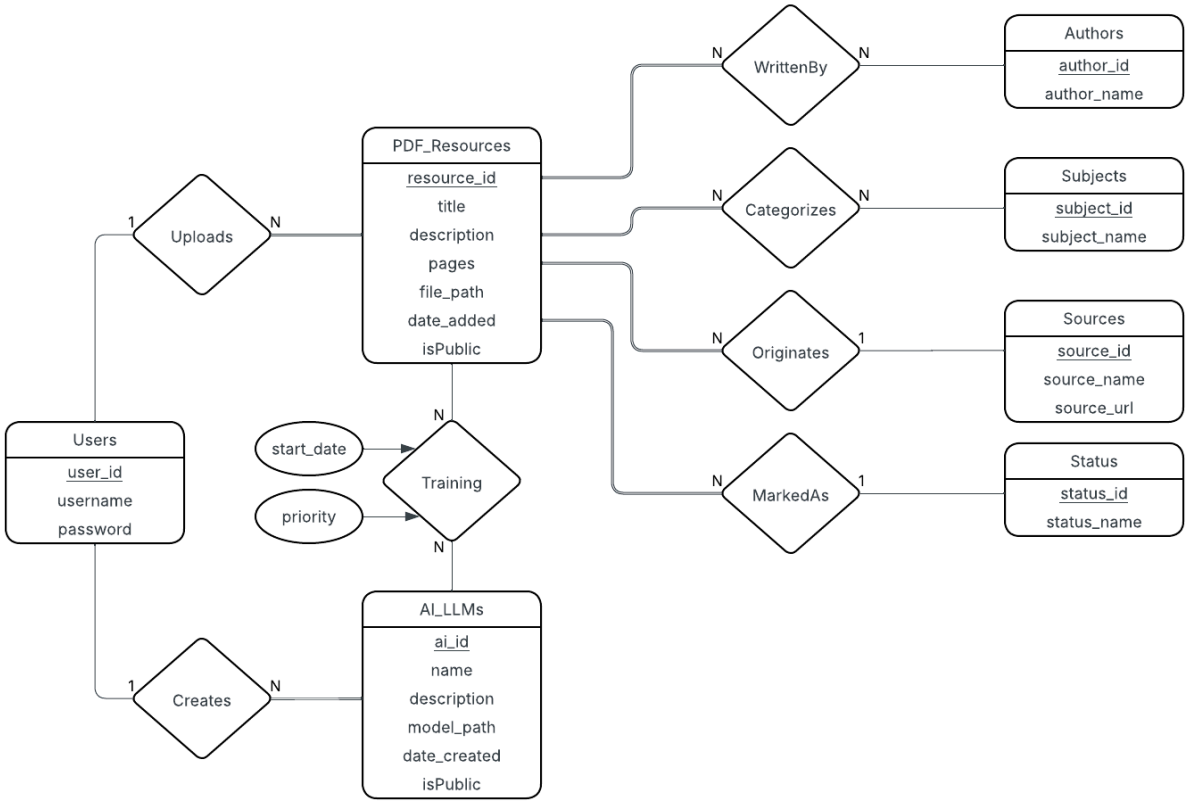


Table Schema

